UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURE AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS
COURSE MANUAL, FIRST SEMESTER, 2022/2023 ACADEMIC YEAR
MARCH, 2022



Course Code/Title: PHY 101 – GENERAL PHYSICS (2 credits)
Course Contents: This course covers in an elementary manner the basic concepts of physics and its applications to real world problems.

This course is intended to dispel the popular notion that physics is only accessible to smart people and also that physics is just a collection of equations used to solve obscure problems. Physics is interesting, useful, and valuable and it is fun and cool when you apply it to everyday life.

It is one of the foundations on which other disciplines like astronomy, chemistry, geology etc. are built. Physics is the most fundamental science and it is concerned with the basic principles of the universe.

General Physics I is the first course of a two semester introduction to general physics, preparing for careers in Biological and Physical sciences, science and mathematics education, Optometry, lab technology and Nursing.

The main topics are Physical Quantities, Vectors, Kinematics, Work, Energy and Power, Fluids, and Thermodynamics.

Our goal is for you to develop an understanding and intuition for physics so that you can solve practical problems. We feel that the only way to accomplish this goal is by thinking about and solving lots of problems and experimenting in the laboratory.

We hope that Physics 101 will develop the critical thinking in collaborative skills you will need in your future career. Key to this is the need to help you gain conceptual understanding of the physical universe by studying the interrelationship between concepts that are the framework of physics and to apply mathematical tools as an aid in understanding those relationships. At the end of this course you would have improved your ability to reason and to think critically.

You are welcome to our course. It is our hope that you will find this manual very useful and appropriate and by sticking to the guidelines in the course schedule it will help you to understand Physics concepts.

PREREQUISITE:
The prerequisites for this course include a) a strong desire to learn concepts in physics and b) a basic understanding of elementary mathematics and physics.

Instructor Information:
Dr. Kwesi Quagraine
Department of Physics
School of Physical Sciences
College of Agriculture and Natural Sciences
University of Cape Coast

Tel. No.: 050 781 2543
Email: kwesi.quagraine@ucc.edu.gh

Teaching Assistant Information:
Ms. Genevieve Nartey
Tel. No.: 055 845 4331

Teaching/Tutorial:
13 weeks 2 hours of lectures per week + 1 hour tutorials

Times:
Thur, 12.30 – 2.30 = lectures
Arrange with TA Genevieve for Tutorials

Grading/Evaluation:
Minimum of 2 quizzes, 2 assignments = 40%
End of semester examinations=60%

Course policies
Attendance: 100% attendance except with written permission from course lecturer.
Late/missing work: Gets Zero for no workdone
Cheating/Plagiarism: Not Allowed

Subject content:

1. **Physical quantities**, SI units, Derived units, conversions, precision and accuracy
2. **Vectors**, scalars, vector components, unit vectors, Graphical methods
3. **Dynamics**, Force, mass, weight, Newton's laws of motion, collisions, inertial frames of reference, friction
4. **Kinematics**, Distance and displacement, speed, velocity, kinematic equations, circular, simple harmonic and projectile motions
5. **Elasticity**, Stress, strain, modulus of elasticity
6. **Work, Energy & Power**, Work done by a variable force, expression for kinetic energy, work-energy theorem
7. **Fluids**, States of matter, hydrostatics, hydrodynamics
8. **Thermodynamics & Kinetic Theory**, Atomic mass unit, Avogadro's number, temperature scales, kinetic theory of gases, mean free path, heat energy, thermal conductivity, heat capacities, The 1st law of thermodynamics

Course Learning Outcomes:
By the end of this course, learners should be able to:

1 Differentiate between SI units and other units
2. Manipulate and appreciate vectors
3. Distinguish between mass and weight
4. Solve kinematic problems
5. Know the states of matter

Course length and pace:
This course is delivered over a 13-week period with two hours of lectures each week.

Academic integrity:
Students may discuss their assignments with each other and work together on them when deemed appropriate. Each student is expected to submit his/her own individual assignment unless they are specifically assigned in

groups to do group work.

Course materials:
Required textbook:
[1] Introduction to General Physics I by S. Y. Mensah et al.
[2] Cutnell, J.D. & Johnson, K. W. (2009). *Physics* 8th Edition. John-Wiley and Sons Inc.
[3] Resnick, R., Halliday, D., & Walker, J. (2013). *Fundamentals of Physics* (10th ed.), John Wiley
[4] Serway, R. A. & Jewett, Jr., J. W. (2014). *Physics for Scientists and Engineers with Modern Physics* (9th ed). Brooks/Cole, USA.

Course schedule

Date/ Week	**Content topics**	**Learning activities**	**Preparation before class**	**After class assignments**
1	Introduction of Course Manual. Dimensional Analysis	Lecturer introduces course manual Students discuss MKS units, standard prefixes, precision and standard figures	Introduction to General Physics I by S. Y. Mensah. Read pages 1 - 23	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 23 - 24
2	Vectors	Lecturer introduces topic and discusses Cartesian coordinates, displacement, addition, parallelogram law with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 25 - 42	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 43 - 44

3	Kinematics	Lecturer introduces topic and discusses Uniformly accelerated motion, Circular motion, Simple harmonic motion, Projectile motion with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 88 - 136	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 137-138
4	Kinematics	Week 3 is continued	Introduction to General Physics by S. Y. Mensah. Read pages 88 - 136	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 137 - 138
5	Dynamics **(Quiz I)**	Lecturer introduces topic and discusses Newton's laws of motion, mass and weight, inertial frames of reference, friction, strings pulleys and inclines with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 45 - 83	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 83 - 87
6	Work, Energy and Power	Lecturer introduces topic and discusses Work, energy, energy conservation, work-energy conversion,	Introduction to General Physics by S. Y. Mensah. Read pages 139 - 164	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 165 - 166

		power with student contributions		

7	Elasticity	Lecturer introduces topic and discusses Strain, stress, Young's modulus, Bulk and shear moduli with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 167-180	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 180 - 182
8	Fluids	Lecturer introduces topic and discusses Hydrostatics, Pascal's principle, Archimedes principle, fluid flow, viscosity, Bernoulli's principle with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 183 - 239	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 240- 241
9		Week 8 is continued	Introduction to General Physics by S. Y. Mensah. Read pages 183 - 239	Solve questions in Introduction to General Physics I by S. Y. Mensah pages 240 - 241

	Fluids			
10	Kinetic theory	Lecturer introduces topic and discusses Avogadro's number, mass of molecule, kinetic theory, average translational ke, root mean square speed, mean free path with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 242 - 270	Solve questions in Introduction to General Physics I by S. Y. Mensah page 284
11	Thermodynamics (Quiz)	Lecturer introduces topic and discusses Heat energy, thermal conductivity, specific heat, adiabatic processes, The 1st law of thermodynamics, work done by a system with student contributions	Introduction to General Physics by S. Y. Mensah. Read pages 270 - 284	Solve questions in Introduction to General Physics I by S. Y. Mensah page 285, questions 8.3 – 8.7